

7.5 Commutators and Trace-Zero Matrices

The expression $AB - BA$ has come up several times already. In Chapter 0 its trace was 0, which showed that $AB - BA = I_n$ is impossible over $\mathbb{R}$ (Exercise (C1: $AB - BA = I$ is impossible, or is it?)). In Chapter 3 the same trace argument showed that the relation $DS - SD = \text{id}$ between differentiation and multiplication by x on $F[x]$ has no finite-dimensional analogue in characteristic 0 (Exercise (C1: The Heisenberg relation)). Both arguments use one direction only: every matrix of the form $AB - BA$ has trace zero. This section asks whether the converse holds. Is every trace-zero matrix of the form $AB - BA$? The answer is yes over a field of characteristic 0, and the proof is a good example of the block techniques of this chapter: move the matrix into a convenient shape by a similarity, one row and column at a time, and then solve an equation that has become entrywise.

Which fields. Commutators, their basic rules, the first similarity lemma and the final theorem on spans work over **every** field F . The lemma on zero diagonals and Shoda's theorem assume that F has **characteristic** 0. We point out exactly where that hypothesis is used, and what happens in characteristic p .

7.5.1 Commutators

Two matrices A and B commute when $AB = BA$, and most pairs do not. The difference $AB - BA$ measures the failure, and it keeps recurring: in the trace arguments above, in the relation between differentiation and multiplication by x , and in the description of trace-like functionals (Exercise (C2: The trace is the only trace-like functional)). So we give it a name.

The commutator of two matrices is the amount by which they fail to commute.

Definition 7.5.1 — Commutator

Let $n \geq 1$ and $A, B \in M_n(F)$. The **commutator** of A and B is

$$[A, B] := AB - BA \in M_n(F).$$

A matrix $C \in M_n(F)$ is a **commutator** if $C = [X, Y]$ for **some** $X, Y \in M_n(F)$. For operators $S, T \in \mathcal{L}(V)$ on a vector space V , we write likewise $[S, T] := ST - TS \in \mathcal{L}(V)$.

In words: $[A, B]$ is “ A then B ” minus “ B then A ”, in that order, and it is the zero matrix exactly when A and B commute. The second sentence of the definition is about a single matrix: being a commutator means that **there exist** two matrices whose commutator it is, and it says nothing about which ones.

Examples.

1. **Degenerate cases.** $[A, A] = 0$ and $[A, cI_n] = cA - cA = 0$ for every $c \in F$. So the zero matrix is a commutator, in many ways. In $M_1(F) = F$ every commutator is $ab - ba = 0$, so for $n = 1$ the only commutator is 0; this is the smallest case, and it already matches the trace condition, since the only 1×1 matrix of trace 0 is (0).
2. **Matrix units.** Recall that $E_{ij}E_{kl}$ is E_{il} if $j = k$ and 0 otherwise (Definition 0.9.9). So

for $i \neq k$,

$$[E_{ij}, E_{jk}] = E_{ik} - 0 = E_{ik}, \quad \text{and} \quad [E_{ij}, E_{ji}] = E_{ii} - E_{jj} \quad (i \neq j).$$

For instance $[E_{12}, E_{21}] = \text{diag}(1, -1)$ in $M_2(F)$, and $[E_{11}, E_{12}] = E_{12}$. Both have trace 0.

3. **Operators on an infinite-dimensional space.** On $F[x]$, with D differentiation and S multiplication by x , Exercise (C1: The Heisenberg relation) (a) shows $[D, S] = \text{id}$. So the identity is a commutator of operators on $F[x]$. This does not contradict anything, because there is no trace on $\mathcal{L}(F[x])$: the trace was defined only on finite-dimensional spaces (Definition 3.10.2).

Non-example by minimal change. Change the minus sign to a plus: $AB + BA$. It is still linear in A and in B , and it is still built from the two products AB and BA . What is lost is the trace: $\text{tr}(AB + BA) = 2 \text{tr}(AB)$, which has no reason to vanish. For example $E_{12}E_{21} + E_{21}E_{12} = E_{11} + E_{22} = I_2$, with trace 2. It also no longer detects commuting: $AA + AA = 2A^2$ is non-zero for $A = I_n$ over $\mathbb{Q}$, although A commutes with itself.

Why this definition. The order in $AB - BA$ is a convention; the other order gives $[B, A] = -[A, B]$, which is a commutator too. The notion is invariant under a change of basis: a similarity applied to X and Y applies to $[X, Y]$. So “being a commutator” is a property of the operator $\mathbf{x} \mapsto C\mathbf{x}$, not of the basis used to write C , and we are free to choose a convenient basis. This is what the proof of the main theorem exploits.

The basic rules are collected here.

Proposition 7.5.1 — Rules for Commutators

Let $A, A', B \in M_n(F)$, $c \in F$, and let $P \in M_n(F)$ be invertible. Then:

- (a) $[A + A', B] = [A, B] + [A', B]$, $[cA, B] = c[A, B]$, and $[B, A] = -[A, B]$; in particular the commutator is linear in each slot.
- (b) $P^{-1}[A, B]P = [P^{-1}AP, P^{-1}BP]$.
- (c) $\text{tr}[A, B] = 0$.

Proof.

- (a) By Theorem 0.9.4, $(A + A')B - B(A + A') = (AB - BA) + (A'B - BA')$ and $(cA)B - B(cA) = c(AB - BA)$. The last identity is $BA - AB = -(AB - BA)$. Linearity in the second slot follows from the first slot and antisymmetry.
- (b) Inserting $PP^{-1} = I_n$, $P^{-1}ABP = (P^{-1}AP)(P^{-1}BP)$, and likewise for BA . Subtract, using distributivity.
- (c) By Theorem 0.9.6 (1) and (3), $\text{tr}(AB - BA) = \text{tr}(AB) - \text{tr}(BA) = 0$.

■

So every commutator lies in the subspace of trace-zero matrices of Example (Trace-Free Matrices). The question of this section is whether it fills that subspace.

7.5.2 The identity in characteristic p

Rule (c) is the whole content of the Heisenberg exercises: if $[X, Y] = I_n$, then $n \cdot 1 = \text{tr } I_n = 0$, which is impossible when F has characteristic 0 (Definition 0.7.2). The obstruction is the number $n \cdot 1$, not the identity matrix itself. In characteristic p , the $p \times p$ identity has trace $p \cdot 1 = 0$, and the obstruction disappears. In fact the pair D, S of Example 3 survives in a finite form.

Example. (The Identity as a Commutator in Characteristic p)

Let F have prime characteristic p (for instance $F = \mathbb{F}_p$). Let $D_p \in M_p(F)$ have $(k, k+1)$ -entry $k \cdot 1$ for $1 \leq k \leq p-1$, and let $S_p \in M_p(F)$ have $(k+1, k)$ -entry 1 for $1 \leq k \leq p-1$, all other entries 0. Show that $[D_p, S_p] = I_p$.

Solution. The columns of these matrices describe differentiation and multiplication by x on the list $1, x, \dots, x^{p-1}$, where x^p is replaced by 0. By Theorem 0.9.2, $S_p \mathbf{e}_k = \mathbf{e}_{k+1}$ for $k < p$ and $S_p \mathbf{e}_p = \mathbf{0}$, while $D_p \mathbf{e}_1 = \mathbf{0}$ and $D_p \mathbf{e}_k = (k-1)\mathbf{e}_{k-1}$ for $k \geq 2$. Hence

$$D_p S_p \mathbf{e}_k = k \mathbf{e}_k \quad (k < p), \quad D_p S_p \mathbf{e}_p = \mathbf{0}, \quad S_p D_p \mathbf{e}_k = (k-1) \mathbf{e}_k \quad (1 \leq k \leq p),$$

where for $k=1$ we used $S_p D_p \mathbf{e}_1 = \mathbf{0} = 0 \mathbf{e}_1$. Subtracting, $[D_p, S_p] \mathbf{e}_k = \mathbf{e}_k$ for $k < p$, and $[D_p, S_p] \mathbf{e}_p = -(p-1)\mathbf{e}_p = (1-p)\mathbf{e}_p = \mathbf{e}_p$, because $p \cdot 1 = 0$ in F (Definition 0.7.2, Lemma 0.7.4). So every column of $[D_p, S_p]$ is the corresponding column of I_p . For $p=2$ this is the pair of Exercise (C1: The Heisenberg relation) (d), transposed.

Warning. “The identity is never a commutator” is a statement about characteristic 0. The trace test says that a commutator has trace 0, and $\text{tr } I_n = n \cdot 1$. Over $\mathbb{R}$ this is $n \neq 0$; over $\mathbb{F}_p$ it is 0 whenever p divides n , and then the test is silent. The example above shows that I_p really is a commutator over $\mathbb{F}_p$. Whenever an argument ends with “ $n \cdot 1 = 0$, contradiction”, check which fields it covers.

7.5.3 Trace-zero matrices with zero diagonal

We want to write a given trace-zero A as $[X, Y]$. Look first at the easiest choice of X , a diagonal matrix $X = \text{diag}(x_1, \dots, x_n)$. For any $Y = (y_{ij})$, Definition 0.9.9 gives $(XY)_{ij} = x_i y_{ij}$ and $(YX)_{ij} = y_{ij} x_j$, so

$$[X, Y]_{ij} = (x_i - x_j) y_{ij}. \quad (*)$$

The diagonal entries of $[X, Y]$ are all 0, and each off-diagonal entry can be anything, provided $x_i \neq x_j$. So a diagonal X with distinct entries produces **exactly** the matrices with zero diagonal. A trace-zero matrix need not have zero diagonal, but by Proposition 7.5.1 (b) we may replace A by any matrix similar to it. The plan is therefore:

- ① find a similarity that makes the diagonal of A zero; ② solve $(*)$ entrywise.

Step ① is the heart of the matter. We do it one diagonal entry at a time, starting with the $(1, 1)$ -entry.

A matrix is **scalar** if it equals λI_n for some $\lambda \in F$. A scalar matrix is similar only to itself, since $P^{-1}(\lambda I_n)P = \lambda I_n$, so its diagonal cannot be changed. Every other matrix can at least be given a zero in the corner. This first lemma holds over every field.

Lemma 7.5.2 — A Non-Scalar Matrix Has a Similar Matrix with Zero Corner

Let F be any field, let $n \geq 2$, and let $A \in M_n(F)$ be **not** scalar. Then there is an invertible $P \in M_n(F)$ such that the first column of $P^{-1}AP$ is $\mathbf{e}_2$. In particular

$$P^{-1}AP = \begin{pmatrix} 0 & \mathbf{r}^T \\ \mathbf{e}_1 & A_1 \end{pmatrix}$$

for some $\mathbf{r} \in F^{n-1}$ and $A_1 \in M_{n-1}(F)$ with $\text{tr } A_1 = \text{tr } A$, where $\mathbf{e}_1 \in F^{n-1}$.

Idea. Work backwards from the goal. The first column of $P^{-1}AP$ is the coordinate vector of $A\mathbf{p}_1$ in the basis formed by the columns $\mathbf{p}_1, \dots, \mathbf{p}_n$ of P . For it to be $\mathbf{e}_2$, we need $A\mathbf{p}_1 = \mathbf{p}_2$. So we want a vector $\mathbf{v}$ such that $\mathbf{v}$ and $A\mathbf{v}$ are independent, and then we build a basis starting $\mathbf{v}, A\mathbf{v}$. Such a $\mathbf{v}$ exists unless A sends every vector to a multiple of itself, and a short argument shows that this forces A to be scalar.

Proof. We first show that there is $\mathbf{v} \in F^n$ with $(\mathbf{v}, A\mathbf{v})$ linearly independent. Suppose not. Then for every $\mathbf{v} \neq \mathbf{0}$ there are $a, b \in F$, not both zero, with $a\mathbf{v} + bA\mathbf{v} = \mathbf{0}$. If $b = 0$, then $a \neq 0$ and $a\mathbf{v} = \mathbf{0}$, which forces $\mathbf{v} = \mathbf{0}$ (Theorem 1.1.9), a contradiction; so $b \neq 0$ and $A\mathbf{v} = \lambda_{\mathbf{v}}\mathbf{v}$ with $\lambda_{\mathbf{v}} = -a/b$. Apply this to $\mathbf{e}_i, \mathbf{e}_j$ and $\mathbf{e}_i + \mathbf{e}_j$ for $i \neq j$:

$$\lambda_{\mathbf{e}_i+\mathbf{e}_j}\mathbf{e}_i + \lambda_{\mathbf{e}_i+\mathbf{e}_j}\mathbf{e}_j = A(\mathbf{e}_i + \mathbf{e}_j) = A\mathbf{e}_i + A\mathbf{e}_j = \lambda_{\mathbf{e}_i}\mathbf{e}_i + \lambda_{\mathbf{e}_j}\mathbf{e}_j.$$

Since $\mathbf{e}_i, \mathbf{e}_j$ are linearly independent, comparing coefficients (Theorem 1.4.3) gives $\lambda_{\mathbf{e}_i} = \lambda_{\mathbf{e}_i+\mathbf{e}_j} = \lambda_{\mathbf{e}_j}$. Hence all $\lambda_{\mathbf{e}_i}$ equal a common λ , so $A\mathbf{e}_i = \lambda\mathbf{e}_i$ for every i , and by Theorem 0.9.2 the columns of A are those of λI_n . So $A = \lambda I_n$ is scalar, contrary to the hypothesis.

Take such a $\mathbf{v}$. By Theorem 1.6.5, the independent list $(\mathbf{v}, A\mathbf{v})$ extends to a basis $(\mathbf{v}, A\mathbf{v}, \mathbf{w}_3, \dots, \mathbf{w}_n)$ of F^n . Let P be the matrix with these columns; it is invertible by Theorem 2.4.4. Then $P\mathbf{e}_1 = \mathbf{v}$ and $P\mathbf{e}_2 = A\mathbf{v}$ (Theorem 0.9.3), so the first column of $P^{-1}AP$ is

$$P^{-1}AP\mathbf{e}_1 = P^{-1}A\mathbf{v} = P^{-1}P\mathbf{e}_2 = \mathbf{e}_2.$$

Partitioning $P^{-1}AP$ after the first row and column gives the displayed form. Finally, $\text{tr}(P^{-1}AP) = \text{tr } A$ by Theorem 3.7.5, and $\text{tr}(P^{-1}AP) = 0 + \text{tr } A_1$ by Definition 0.9.12, so $\text{tr } A_1 = \text{tr } A$. ■

Quick check. Over $\mathbb{Q}$, let $A = \begin{pmatrix} 2 & 1 \\ 0 & -2 \end{pmatrix}$. Using $\mathbf{v} = \mathbf{e}_2$ in the proof of Lemma 7.5.2, find P and $P^{-1}AP$, and check that its diagonal is zero.

Solution. $A\mathbf{e}_2 = (1, -2)$, which is not a multiple of $\mathbf{e}_2$, so $(\mathbf{e}_2, A\mathbf{e}_2)$ is independent and already a basis of $\mathbb{Q}^2$. Then $P = \begin{pmatrix} 0 & 1 \\ 1 & -2 \end{pmatrix}$, with $\det P = -1$ and $P^{-1} = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}$.

We get $AP = \begin{pmatrix} 1 & 0 \\ -2 & 4 \end{pmatrix}$ and $P^{-1}AP = \begin{pmatrix} 0 & 4 \\ 1 & 0 \end{pmatrix}$. The first column is $\mathbf{e}_2$, as the lemma promises; the $(2, 2)$ -entry is 0 as well, because the trace is 0 and trace is a similarity invariant.

The lemma controls one diagonal entry and hands us a smaller matrix A_1 with the same trace. That is exactly the setting for induction on the size. The only thing that can stop the induction is a scalar A_1 , and here the trace saves us, provided the field has characteristic 0.

Lemma 7.5.3 — Trace Zero Means Similar to Zero Diagonal

Let F be a field of **characteristic** 0, let $n \geq 1$, and let $A \in M_n(F)$ with $\text{tr } A = 0$. Then A is similar to a matrix all of whose diagonal entries are 0.

Idea. Induction on n . If A is scalar, $A = \lambda I_n$ with $n\lambda = 0$, and characteristic 0 forces $\lambda = 0$: then $A = 0$ already has zero diagonal. Otherwise, Lemma 7.5.2 puts a 0 in the corner and leaves a trace-zero block A_1 of size $n-1$. The induction hypothesis conjugates A_1 by some Q into zero diagonal. Conjugating the whole matrix by the block diagonal matrix $(1) \oplus Q$ does this to the lower block without disturbing the corner.

Proof. We use induction on n (Theorem 0.2.7). For $n = 1$, $A = (\text{tr } A) = (0)$, which has zero diagonal.

Let $n \geq 2$, and assume the statement for trace-zero matrices of size $n-1$.

Case 1: A is scalar, say $A = \lambda I_n$. Then $0 = \text{tr } A = n \cdot \lambda = (n \cdot 1)\lambda$ by Definition 0.9.12 and Lemma 0.7.4. Since F has characteristic 0 and $n \geq 1$, $n \cdot 1 \neq 0$ (Definition 0.7.2), so $\lambda = 0$. Hence $A = 0$, which has zero diagonal and is similar to itself.

Case 2: A is not scalar. By Lemma 7.5.2 there is an invertible P with

$$M := P^{-1}AP = \begin{pmatrix} 0 & \mathbf{r}^\top \\ \mathbf{e}_1 & A_1 \end{pmatrix}, \quad \text{tr } A_1 = \text{tr } A = 0.$$

By the induction hypothesis there is an invertible $Q \in M_{n-1}(F)$ such that $Z_1 := Q^{-1}A_1Q$ has zero diagonal. Let $R = (1) \oplus Q = \begin{pmatrix} 1 & \mathbf{0}^\top \\ \mathbf{0} & Q \end{pmatrix}$. By Theorem 7.1.4, R is invertible with $R^{-1} = (1) \oplus Q^{-1}$. By Theorem 7.1.1,

$$R^{-1}MR = \begin{pmatrix} 1 & \mathbf{0}^\top \\ \mathbf{0} & Q^{-1} \end{pmatrix} \begin{pmatrix} 0 & \mathbf{r}^\top \\ \mathbf{e}_1 & A_1 \end{pmatrix} \begin{pmatrix} 1 & \mathbf{0}^\top \\ \mathbf{0} & Q \end{pmatrix} = \begin{pmatrix} 0 & \mathbf{r}^\top Q \\ Q^{-1}\mathbf{e}_1 & Q^{-1}A_1Q \end{pmatrix} = \begin{pmatrix} 0 & \mathbf{r}^\top Q \\ Q^{-1}\mathbf{e}_1 & Z_1 \end{pmatrix}.$$

Its $(1, 1)$ -entry is 0, and its other diagonal entries are those of Z_1 , which are 0. Since $R^{-1}MR = (PR)^{-1}A(PR)$ by Theorem 0.9.7 (3), and PR is invertible, A is similar to a matrix with zero diagonal. This completes the induction. ■

Remark. The characteristic was used only in Case 1, and only through $k \cdot 1 \neq 0$ for the sizes $k = 2, \dots, n$ that the induction passes through. So the lemma holds, with the same proof, whenever F has characteristic 0 or characteristic $p > n$. Some hypothesis on the characteristic is needed: for $n = p$, over $\mathbb{F}_p$, the matrix I_p has trace $p \cdot 1 = 0$, but it is

scalar, so it is similar only to itself, and its diagonal entries are 1. The same happens for λI_n with $\lambda \neq 0$ whenever p divides n .

7.5.4 Shoda's theorem

With a zero diagonal available, the diagonal choice of X in $(*)$ finishes the job. All we need are n distinct scalars for the diagonal of X , and characteristic 0 provides them.

Theorem 7.5.4 — Shoda's Theorem

Let F be a field of **characteristic** 0, let $n \geq 1$, and let $A \in M_n(F)$. Then A is a commutator, that is $A = [X, Y]$ for some $X, Y \in M_n(F)$, if and only if $\text{tr } A = 0$.

Idea. One direction is [Proposition 7.5.1 \(c\)](#). For the other: ① by [Lemma 7.5.3](#), $A = PZP^{-1}$ with Z of zero diagonal; ② take $X_0 = \text{diag}(0, 1, \dots, n-1)$, whose entries are distinct in characteristic 0; ③ solve $[X_0, Y_0] = Z$ entrywise with $(*)$, dividing by $x_i - x_j \neq 0$; ④ conjugate back with [Proposition 7.5.1 \(b\)](#).

Proof. $(\Rightarrow)$ If $A = [X, Y]$, then $\text{tr } A = 0$ by [Proposition 7.5.1 \(c\)](#).

$(\Leftarrow)$ Suppose $\text{tr } A = 0$. By [Lemma 7.5.3](#) there is an invertible P such that $Z = (z_{ij}) := P^{-1}AP$ has $z_{ii} = 0$ for all i .

For $1 \leq i \leq n$ let $x_i := (i-1) \cdot 1 \in F$, and $X_0 := \text{diag}(x_1, \dots, x_n)$. For $i < j$, [Lemma 0.7.4](#) gives $x_j - x_i = (j-i) \cdot 1$, which is non-zero because $j-i \geq 1$ and F has characteristic 0 ([Definition 0.7.2](#)). So the x_i are distinct, and we may define $Y_0 = (y_{ij}) \in M_n(F)$ by

$$y_{ij} := \frac{z_{ij}}{x_i - x_j} \quad (i \neq j), \quad y_{ii} := 0.$$

By $(*)$, $[X_0, Y_0]_{ij} = (x_i - x_j)y_{ij} = z_{ij}$ for $i \neq j$, and $[X_0, Y_0]_{ii} = 0 = z_{ii}$. Hence $[X_0, Y_0] = Z$.

Let $X = PX_0P^{-1}$ and $Y = PY_0P^{-1}$. Applying [Proposition 7.5.1 \(b\)](#) with the invertible matrix P^{-1} ,

$$[X, Y] = [PX_0P^{-1}, PY_0P^{-1}] = P[X_0, Y_0]P^{-1} = PZP^{-1} = A.$$

This proves that A is a commutator. ■

The proof used characteristic 0 twice: in the lemma, to rule out non-zero scalar blocks of trace zero, and here, to find n distinct diagonal entries. The second use needs only $(j-i) \cdot 1 \neq 0$ for $1 \leq j-i \leq n-1$, and with the remark after [Lemma 7.5.3](#), the same proof works whenever the characteristic of F is 0 or larger than n . Neither use is essential for the **theorem**. In fact the statement “a square matrix over a field is a commutator if and only if its trace is 0” is true over **every** field. The proof in positive characteristic needs a different normal form and more work, and we do not give it. [Example \(The Identity as a Commutator in Characteristic \$p\$ \)](#) is a small instance: I_p over $\mathbb{F}_p$ is a commutator, although [Lemma 7.5.3](#) fails for it.

The proof is constructive, and it is worth running once in full.

Example. (Writing a Trace-Zero 3×3 Matrix as a Commutator)

Over $\mathbb{Q}$, write $A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & -2 \end{pmatrix}$ as a commutator $[X, Y]$.

Solution. The trace is $1 + 1 - 2 = 0$. We follow the proofs step by step.

First corner. A is not scalar. Take $\mathbf{v} = \mathbf{e}_1$; then $A\mathbf{v} = (1, 0, 1)$ is not a multiple of $\mathbf{e}_1$. Extend $(\mathbf{e}_1, A\mathbf{e}_1)$ by $\mathbf{e}_2$, which is not in their span, to a basis, and let

$$P_1 = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad M = P_1^{-1}AP_1 = \begin{pmatrix} 0 & 2 & 2 \\ 1 & -1 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

(To check M without inverting, verify $P_1M = AP_1$: both equal $\begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 1 & -1 & 0 \end{pmatrix}$.) The first column is $\mathbf{e}_2$, and the lower block is $A_1 = \begin{pmatrix} -1 & 0 \\ 1 & 1 \end{pmatrix}$, with trace 0.

Second corner. A_1 is not scalar. With $\mathbf{w} = \mathbf{e}_1 \in \mathbb{Q}^2$, $A_1\mathbf{w} = (-1, 1)$ is independent of $\mathbf{e}_1$, so

$$Q = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}, \quad Q^{-1} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad Z_1 = Q^{-1}A_1Q = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Assemble. With $\mathbf{r}^\top = (2, 2)$, the proof of [Lemma 7.5.3](#) gives $\mathbf{r}^\top Q = (2, 0)$ and $Q^{-1}\mathbf{e}_1 = \mathbf{e}_1$, so for $P = P_1((1) \oplus Q)$,

$$P = \begin{pmatrix} 1 & 1 & -1 \\ 0 & 0 & 1 \\ 0 & 1 & -1 \end{pmatrix}, \quad Z = P^{-1}AP = \begin{pmatrix} 0 & 2 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad P^{-1} = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

Solve entrywise. With $X_0 = \text{diag}(0, 1, 2)$, the rule $y_{ij} = z_{ij}/(x_i - x_j)$ gives $y_{12} = \frac{2}{-1} = -2$, $y_{21} = \frac{1}{1} = 1$, $y_{23} = \frac{1}{-1} = -1$, $y_{32} = \frac{1}{1} = 1$, and $y_{13} = y_{31} = 0$:

$$Y_0 = \begin{pmatrix} 0 & -2 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}.$$

Conjugate back. $X = PX_0P^{-1}$ and $Y = PY_0P^{-1}$ are

$$X = \begin{pmatrix} 0 & -1 & 1 \\ 0 & 2 & 0 \\ 0 & -1 & 1 \end{pmatrix}, \quad Y = \begin{pmatrix} 1 & -4 & -4 \\ 0 & 1 & 1 \\ 1 & -2 & -2 \end{pmatrix}.$$

$$\text{Check: } XY = \begin{pmatrix} 1 & -3 & -3 \\ 0 & 2 & 2 \\ 1 & -3 & -3 \end{pmatrix} \text{ and } YX = \begin{pmatrix} 0 & -5 & -3 \\ 0 & 1 & 1 \\ 0 & -3 & -1 \end{pmatrix}, \text{ and } XY - YX = A.$$

Nothing about these X and Y is unique. Adding to Y any matrix that commutes with X , for instance a polynomial in X , gives another solution, by [Proposition 7.5.1](#) (a).

7.5.5 The span of the commutators

Shoda's theorem is about single commutators. A weaker question, with an easier answer that holds over **every** field, is which matrices are **sums** of commutators. Since the trace-zero matrices form a subspace and contain all commutators, the span of the commutators lies inside it. Matrix units show that it is all of it.

Theorem 7.5.5 — Commutators Span the Trace-Zero Matrices

Let F be any field and $n \geq 1$. The span of all commutators $[X, Y]$ with $X, Y \in M_n(F)$ is the subspace $\{A \in M_n(F) : \text{tr } A = 0\}$. More precisely, every trace-zero matrix is a linear combination of the commutators

$$E_{ij} = [E_{ii}, E_{ij}] \quad (i \neq j) \quad \text{and} \quad E_{ii} - E_{nn} = [E_{in}, E_{ni}] \quad (1 \leq i \leq n-1).$$

Proof. Let $W = \{A \in M_n(F) : \text{tr } A = 0\}$, a subspace by Example (Trace-Free Matrices). By [Proposition 7.5.1](#) (c), every commutator lies in W , so their span is contained in W (Theorem 1.3.1, the span being the smallest subspace containing them).

For the reverse inclusion, the displayed identities hold by Example 2 after [Definition 7.5.1](#): for $i \neq j$, $[E_{ii}, E_{ij}] = E_{ij}$, and for $i < n$, $[E_{in}, E_{ni}] = E_{ii} - E_{nn}$. Now let $A = (a_{ij}) \in W$. Writing A in the basis of matrix units (Example (Standard Bases)) and using $a_{nn} = -\sum_{i < n} a_{ii}$,

$$A = \sum_{i \neq j} a_{ij} E_{ij} + \sum_{i=1}^{n-1} a_{ii} E_{ii} + a_{nn} E_{nn} = \sum_{i \neq j} a_{ij} E_{ij} + \sum_{i=1}^{n-1} a_{ii} (E_{ii} - E_{nn}).$$

So A is a linear combination of commutators, hence lies in their span. (For $n = 1$, $W = \{0\}$ and both sides are $\{0\}$.)

■

The theorem lists $n(n-1) + (n-1) = n^2 - 1$ commutators spanning the trace-zero matrices. Since $\text{tr}: M_n(F) \rightarrow F$ is onto ($\text{tr}(cE_{11}) = c$), the Rank–Nullity Theorem (Theorem 3.3.1) gives that its kernel has dimension $n^2 - 1$, so by Theorem 1.6.4 (b) these commutators form a basis of it. Over a field of characteristic 0, Shoda's theorem says much more: the sum of commutators produced here can always be collapsed into a single commutator. In general the set of commutators is not obviously closed under addition, and that is exactly what makes Shoda's theorem a theorem.

Warning. The span of a set is not the set. [Theorem 7.5.5](#) shows that every trace-zero matrix is a **sum** of commutators, over any field, by a two-line argument. It does not show that every trace-zero matrix **is** a commutator. For comparison, in $M_2(F)$ the

matrices of rank at most 1 span everything (the four matrix units have rank 1), yet I_2 has rank 2: a set can span a subspace without filling it. For commutators it happens that the set does fill the subspace, but the proof needs the similarity argument of [Lemma 7.5.3](#), not just matrix units.

The span theorem gives a clean proof of a fact from Chapter 3: a linear functional φ on $M_n(F)$ with $\varphi(AB) = \varphi(BA)$ is a multiple of the trace. Such a φ kills every commutator, hence their span, which is the kernel of the trace; [Exercise \(C1: Trace-like functionals, again\)](#) below completes the argument.

7.5.6 Exercises

7.5.6.1 A. Check your understanding

Exercise A1.

- Define the commutator $[A, B]$, and state what it means for a matrix to be a commutator.
- State Shoda's theorem, including the hypothesis on the field.
- True or false: $[A, B] = [B, A]$ for all $A, B \in M_n(F)$. Justify your answer.
- True or false: over every field, every trace-zero matrix is similar to a matrix with zero diagonal. Justify your answer.
- Name the two places where the proof of Shoda's theorem uses characteristic 0.
- Explain why [Theorem 7.5.5](#) does not by itself prove Shoda's theorem.

Solution.

- $[A, B] = AB - BA$ ([Definition 7.5.1](#)). A matrix $C \in M_n(F)$ is a commutator if $C = [X, Y]$ for some $X, Y \in M_n(F)$.
- If F has characteristic 0 and $A \in M_n(F)$, then $A = [X, Y]$ for some $X, Y \in M_n(F)$ if and only if $\text{tr } A = 0$ ([Theorem 7.5.4](#)).
- False. $[B, A] = -[A, B]$ ([Proposition 7.5.1](#) (a)), so equality holds only when $[A, B] = 0$. For example $[E_{12}, E_{21}] = \text{diag}(1, -1)$ and $[E_{21}, E_{12}] = \text{diag}(-1, 1)$ over $\mathbb{Q}$.
- False. Over $\mathbb{F}_2$, I_2 has trace $1 + 1 = 0$, but it is scalar, so every matrix similar to it equals $P^{-1}I_2P = I_2$, whose diagonal entries are 1.
- In [Lemma 7.5.3](#), to conclude from $(n \cdot 1)\lambda = 0$ that a trace-zero scalar matrix λI_n is 0; and in [Theorem 7.5.4](#), to find n distinct scalars $x_i = (i - 1) \cdot 1$ so that we can divide by $x_i - x_j$.
- It says every trace-zero matrix is a **linear combination** of commutators. Shoda's theorem says it is a **single** commutator. A sum of commutators need not visibly be a commutator, since $[X, Y] + [X', Y']$ has no evident expression as $[X'', Y'']$.

7.5.6.2 B. Practice

Exercise B1: A 2×2 commutator.

Over $\mathbb{Q}$, let $A = \begin{pmatrix} 3 & -2 \\ 1 & -3 \end{pmatrix}$.

- (a) Using $\mathbf{v} = \mathbf{e}_1$ as in [Lemma 7.5.2](#), find an invertible P such that $Z = P^{-1}AP$ has zero diagonal.
- (b) With $X_0 = \text{diag}(0, 1)$, find Y_0 with $[X_0, Y_0] = Z$.
- (c) Hence write $A = [X, Y]$, and check your answer by multiplying out.

Solution.

- (a) $\text{tr } A = 0$, and $A\mathbf{e}_1 = (3, 1)$ is not a multiple of $\mathbf{e}_1$, so $(\mathbf{e}_1, A\mathbf{e}_1)$ is a basis of $\mathbb{Q}^2$. Let $P = \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}$, so $P^{-1} = \begin{pmatrix} 1 & -3 \\ 0 & 1 \end{pmatrix}$. Then $AP = \begin{pmatrix} 3 & 7 \\ 1 & 0 \end{pmatrix}$ and

$$Z = P^{-1}AP = \begin{pmatrix} 0 & 7 \\ 1 & 0 \end{pmatrix}.$$

The first column is $\mathbf{e}_2$ by [Lemma 7.5.2](#), and the $(2, 2)$ -entry is 0 because the trace is 0.

- (b) By (*), $[X_0, Y]_{12} = (0 - 1)y_{12}$ and $[X_0, Y]_{21} = (1 - 0)y_{21}$, while the diagonal of $[X_0, Y]$ is zero. So $y_{12} = -7$, $y_{21} = 1$, and we may take $y_{11} = y_{22} = 0$: $Y_0 = \begin{pmatrix} 0 & -7 \\ 1 & 0 \end{pmatrix}$.

- (c) By the proof of [Theorem 7.5.4](#), $X = PX_0P^{-1}$ and $Y = PY_0P^{-1}$:

$$X = \begin{pmatrix} 0 & 3 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -3 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 3 \\ 0 & 1 \end{pmatrix}, \quad Y = \begin{pmatrix} 3 & -7 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & -3 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 3 & -16 \\ 1 & -3 \end{pmatrix},$$

using $PX_0 = \begin{pmatrix} 0 & 3 \\ 0 & 1 \end{pmatrix}$ and $PY_0 = \begin{pmatrix} 3 & -7 \\ 1 & 0 \end{pmatrix}$. Check: $XY = \begin{pmatrix} 3 & -9 \\ 1 & -3 \end{pmatrix}$, $YX = \begin{pmatrix} 0 & -7 \\ 0 & 0 \end{pmatrix}$, and $XY - YX = \begin{pmatrix} 3 & -2 \\ 1 & -3 \end{pmatrix} = A$.

Exercise B2: A commutator over $\mathbb{F}_2$.

Work over $\mathbb{F}_2$, and let $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.

- (a) Show that $\text{tr } A = 0$ and that A is not scalar.
- (b) Explain why $\mathbf{v} = \mathbf{e}_1$ cannot be used in [Lemma 7.5.2](#), and use $\mathbf{v} = \mathbf{e}_2$ instead to find P with $P^{-1}AP$ of zero diagonal.
- (c) The field $\mathbb{F}_2$ has exactly two elements. Use them as the diagonal of X_0 to write A as a commutator, by hand.
- (d) Explain why the same method cannot write $I_2 \in M_2(\mathbb{F}_2)$ as a commutator, and recall a pair X, Y that does.

Solution.

- (a) $\text{tr } A = 1 + 1 = 0$ in $\mathbb{F}_2$. A has a non-zero off-diagonal entry, while every scalar matrix has zero off-diagonal entries.
- (b) $A\mathbf{e}_1 = \mathbf{e}_1$, so $(\mathbf{e}_1, A\mathbf{e}_1)$ is dependent. With $\mathbf{v} = \mathbf{e}_2$, $A\mathbf{e}_2 = (1, 1)$, which is not a multiple of $\mathbf{e}_2$, so $(\mathbf{e}_2, A\mathbf{e}_2)$ is a basis. Let $P = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$; since $\det P = -1 = 1$, $P^{-1} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ (check: $PP^{-1} = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} = I_2$ in $\mathbb{F}_2$). Then $AP = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$ and $Z = P^{-1}AP = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.
- (c) Take $X_0 = \text{diag}(0, 1)$. By (*), $[X_0, Y]_{12} = (0 - 1)y_{12} = y_{12}$ (as $-1 = 1$) and $[X_0, Y]_{21} = y_{21}$. So $Y_0 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ gives $[X_0, Y_0] = Z$. Conjugating back, $PX_0 = \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}$ and $PY_0 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$, so

$$X = PX_0P^{-1} = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}, \quad Y = PY_0P^{-1} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Check: $XY = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, $YX = \begin{pmatrix} 2 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$, so $[X, Y] = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = A$. (Here $Y = A$: indeed $[X, A] = A$.)

- (d) The method writes $Z = P^{-1}I_2P$ as $[X_0, Y_0]$ with X_0 diagonal, and by (*) such a commutator has zero diagonal. But I_2 is scalar, so $P^{-1}I_2P = I_2$ for every invertible P , and its diagonal entries are 1. So the method cannot start. Nevertheless I_2 is a commutator: with $X = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ and $Y = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $[X, Y] = \text{diag}(-1, 1) = I_2$ in $\mathbb{F}_2$ (Exercise (C1: The Heisenberg relation) (d), or [Example \(The Identity as a Commutator in Characteristic \$p\$ \)](#) with $p = 2$).

Exercise B3: Commuting with a diagonal matrix.

Let F be any field, let $d_1, \dots, d_n \in F$ be **distinct**, let $D = \text{diag}(d_1, \dots, d_n)$, and define $L: M_n(F) \rightarrow M_n(F)$ by $L(X) = [D, X]$.

- (a) Show that L is linear.
- (b) Prove that $\ker L$ is the set of diagonal matrices and $\text{im } L$ is the set of matrices with zero diagonal.
- (c) Check that your answer to (b) agrees with the Rank–Nullity Theorem.
- (d) Hence decide whether $\begin{pmatrix} 0 & 1 & 2 \\ 3 & 0 & 4 \\ 5 & 6 & 0 \end{pmatrix} \in M_3(\mathbb{Q})$ is a commutator with $X = \text{diag}(1, 2, 3)$, and if so find Y .

Solution.

- (a) By [Proposition 7.5.1](#) (a), $[D, X + X'] = [D, X] + [D, X']$ and $[D, cX] = c[D, X]$.
- (b) By (*), $L(X)_{ij} = (d_i - d_j)x_{ij}$. Since $d_i - d_j \neq 0$ for $i \neq j$, $L(X) = 0$ if and only

if $x_{ij} = 0$ for all $i \neq j$, that is, X is diagonal. Every $L(X)$ has zero diagonal. Conversely, if Z has zero diagonal, the matrix Y with $y_{ij} = z_{ij}/(d_i - d_j)$ for $i \neq j$ and $y_{ii} = 0$ satisfies $L(Y) = Z$. So $\text{im } L$ is exactly the set of zero-diagonal matrices.

- (c) The diagonal matrices have basis $E_{11}, \dots, E_{nn}$, so nullity $L = n$. The zero-diagonal matrices have basis E_{ij} ($i \neq j$), so $\text{rank } L = n^2 - n$. Their sum is $n^2 = \dim M_n(F)$, as Theorem 3.3.1 requires.
- (d) Yes, since the matrix has zero diagonal and 1, 2, 3 are distinct. With $d = (1, 2, 3)$, $y_{ij} = z_{ij}/(d_i - d_j)$ gives

$$Y = \begin{pmatrix} 0 & \frac{1}{-1} & \frac{2}{-2} \\ \frac{3}{1} & 0 & \frac{4}{-1} \\ \frac{5}{2} & \frac{6}{1} & 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 & -1 \\ 3 & 0 & -4 \\ \frac{5}{2} & 6 & 0 \end{pmatrix},$$

and by (b), $[\text{diag}(1, 2, 3), Y]$ is the given matrix.

7.5.6.3 C. Going deeper

Exercise C1: Trace-like functionals, again.

Let F be any field, $n \geq 1$, and let $\varphi: M_n(F) \rightarrow F$ be linear with $\varphi(AB) = \varphi(BA)$ for all $A, B \in M_n(F)$.

- (a) Prove that φ vanishes on every trace-zero matrix.
- (b) Deduce that $\varphi = c \text{ tr}$, where $c = \varphi(E_{11})$.
- (c) Suppose F has characteristic 0 and, in addition, $\varphi(I_n) = n \cdot 1$. Deduce that $\varphi = \text{tr}$.
- (d) Show that the conclusion of (c) fails over $\mathbb{F}_p$ when $n = p$.

Hint: for (a), use Theorem 7.5.5.

Solution.

- (a) For all X, Y , $\varphi([X, Y]) = \varphi(XY) - \varphi(YX) = 0$ by linearity and the hypothesis. So the kernel of φ , a subspace, contains every commutator, hence their span (Theorem 1.3.1), which is the set of trace-zero matrices by Theorem 7.5.5.
- (b) Let $A \in M_n(F)$. Since $\text{tr } E_{11} = 1$, the matrix $A - (\text{tr } A)E_{11}$ has trace 0 (Theorem 0.9.6 (1)), so by (a) and linearity,

$$0 = \varphi(A - (\text{tr } A)E_{11}) = \varphi(A) - (\text{tr } A) \varphi(E_{11}).$$

Hence $\varphi(A) = c \text{ tr } A$ with $c = \varphi(E_{11})$. This recovers Exercise (C2: The trace is the only trace-like functional) by a different route: the matrix-unit computations are now packaged in Theorem 7.5.5.

- (c) By (b), $n \cdot 1 = \varphi(I_n) = c \text{ tr } I_n = c(n \cdot 1)$. Since F has characteristic 0, $n \cdot 1 \neq 0$, so $c = 1$ and $\varphi = \text{tr}$.
- (d) Over $\mathbb{F}_p$ with $n = p$, take $\varphi = 0$. It is linear, satisfies $\varphi(AB) = 0 = \varphi(BA)$, and $\varphi(I_p) = 0 = p \cdot 1$. But $\varphi \neq \text{tr}$, since $\text{tr } E_{11} = 1$. The step that fails is canceling $n \cdot 1 = 0$ in (c).

Exercise C2: When A commutes with $[A, B]$

Let F be a field of characteristic 0, let $A, B \in M_n(F)$, put $C = [A, B]$, and suppose that $AC = CA$. The goal is to prove that C is nilpotent, that is, $C^s = 0$ for some $s \geq 1$.

- (a) Prove that $C^k = [A, BC^{k-1}]$ for every $k \geq 1$. Deduce that $\text{tr } C^k = 0$ for every $k \geq 1$.
- (b) Show that there are an integer $s \geq 1$ and $g \in F[x]$ with $g(0) \neq 0$ such that $C^s g(C) = 0$.
- (c) Let $U = \text{col}(C^s)$ and $r = \dim U$, and suppose $r \geq 1$. Extend a basis of U to a basis of F^n , and let P be the matrix with these columns. Show that

$$P^{-1}CP = \begin{pmatrix} D & * \\ 0 & N \end{pmatrix}, \quad D \in M_r(F),$$

with $N^s = 0$ and $g(D) = 0$.

- (d) Write $g = g_0 + xh$ with $g_0 = g(0) \neq 0$. Show that I_r is a linear combination of $D^s, D^{s+1}, \dots, D^d$ for some $d \geq s$, and use (a) to reach a contradiction.
- (e) Conclude that $C^s = 0$.

Hint: in (c), compare the columns of $P^{-1}C^kP$ for $k = 1$ and $k = s$ with the coordinates of vectors of U ; in (d), take traces.

Solution.

- (a) First, $C^{k-1}A = AC^{k-1}$ for all $k \geq 1$, by induction on k from $CA = AC$. Hence

$$[A, BC^{k-1}] = ABC^{k-1} - BC^{k-1}A = ABC^{k-1} - BAC^{k-1} = (AB - BA)C^{k-1} = C^k.$$

By Proposition 7.5.1 (c), $\text{tr } C^k = 0$.

- (b) By Theorem 5.7.2 there is a non-zero $f \in F[x]$ with $f(C) = 0$. Let $t \geq 0$ be the largest integer with x^t dividing f , and write $f = x^t g$; then $x \nmid g$, so $g(0) \neq 0$ by Theorem 5.1.3. Put $s = t + 1 \geq 1$. By Theorem 0.9.9, $C^s g(C) = C f(C) = 0$.

- (c) Let P have columns $\mathbf{p}_1, \dots, \mathbf{p}_n$, where $(\mathbf{p}_1, \dots, \mathbf{p}_r)$ is a basis of U ; P is invertible by Theorem 2.4.4. For any $k \geq 1$, column j of $P^{-1}C^kP$ is $P^{-1}(C^k \mathbf{p}_j)$, the vector $\mathbf{c}$ with $P\mathbf{c} = C^k \mathbf{p}_j$, that is, the coefficients of $C^k \mathbf{p}_j$ in terms of $\mathbf{p}_1, \dots, \mathbf{p}_n$ (Theorem 0.9.3). If $\mathbf{y} \in U$, then $\mathbf{y} = C^s \mathbf{z}$ and $C\mathbf{y} = C^s(C\mathbf{z}) \in U$, so C maps U into U ; hence for $j \leq r$ the vector $C\mathbf{p}_j$ is a combination of $\mathbf{p}_1, \dots, \mathbf{p}_r$, and column j of $P^{-1}CP$ has zeros below row r . This gives the block form. By Theorem 7.1.1 and induction on k , $(P^{-1}CP)^k = \begin{pmatrix} D^k & * \\ 0 & N^k \end{pmatrix}$, and by Proposition 3.7.6 (c) this is

$$P^{-1}C^kP; \text{ more generally } P^{-1}q(C)P = \begin{pmatrix} q(D) & * \\ 0 & q(N) \end{pmatrix} \text{ for every } q \in F[x].$$

For $k = s$, every vector $C^s \mathbf{p}_j$ lies in U , so every column of $P^{-1}C^sP$ is zero below row r , and $N^s = 0$. For $q = g$: for $j \leq r$, $\mathbf{p}_j \in U$ is $C^s \mathbf{z}$ for some $\mathbf{z}$, so $g(C)\mathbf{p}_j = g(C)C^s \mathbf{z} = \mathbf{0}$ by (b) and Theorem 0.9.9. So the first r columns of $P^{-1}g(C)P$ are zero, and in particular $g(D) = 0$.

- (d) From $g(D) = g_0 I_r + Dh(D) = 0$ we get $I_r = Dh_1(D)$ with $h_1 = -g_0^{-1}h$. Since

polynomials in D commute (Theorem 0.9.9),

$$I_r = (D h_1(D))^s = D^s h_1(D)^s = \sum_{k=s}^d c_k D^k$$

for some $c_k \in F$, as every term of $x^s h_1(x)^s$ has degree at least s . For $k \geq s$, $N^k = N^s N^{k-s} = 0$, so by (c), Theorem 3.7.5 and Definition 0.9.12, $\text{tr } C^k = \text{tr}(P^{-1} C^k P) = \text{tr } D^k + \text{tr } N^k = \text{tr } D^k$. Taking traces,

$$r \cdot 1 = \text{tr } I_r = \sum_{k=s}^d c_k \text{tr } D^k = \sum_{k=s}^d c_k \text{tr } C^k = 0$$

by (a). But $r \geq 1$ and F has characteristic 0, so $r \cdot 1 \neq 0$, a contradiction.

- (e) By (d), $r = 0$, so $\text{col}(C^s) = \{\mathbf{0}\}$ and $C^s = 0$. For example, with $A = E_{12} + E_{23}$ and $B = \text{diag}(0, 1, 2)$ in $M_3(\mathbb{Q})$, $C = [A, B] = E_{12} + E_{23} = A$, which commutes with A , and indeed $C^3 = 0$. In Chapter 8 the statement “ $\text{tr } C^k = 0$ for all $k \geq 1$ ” forces nilpotency in characteristic 0” becomes a statement about eigenvalues.